

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-10

Course: EEE F311 Communication Systems
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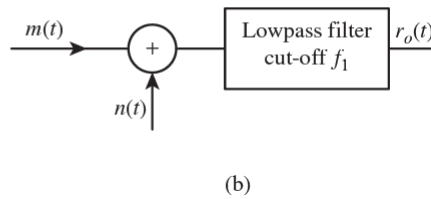
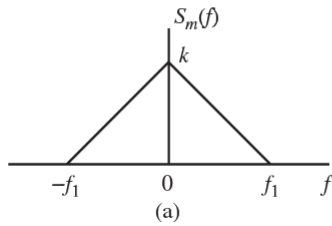
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1 Objectives

- Random Process

2 Tutorial Problems

1. The PSD of signal $m(t)$ is given in the figure. Find the SNR at the output if the noise PSD is $N/2$ (two-sided).



2. Find the PSD of $X(t) + Y(t)$, where $X(t)$ and $Y(t)$ are independent and $X(t) \sim U(-1, 1)$, $Y(t) \sim U(0, 1)$.
3. Q3

The random process $V(t)$ is defined as

$$V(t) = X \cos 2\pi f_c t - Y \sin 2\pi f_c t$$

where X and Y are random variables. Show that $V(t)$ is wide-sense stationary if and only if $E(X) = E(Y) = 0$, $E(X^2) = E(Y^2)$, and $E(XY) = 0$.

1. Solution is appended

(a) The lowpass filter allows the signal to pass through it but cuts off the noise spectrum beyond f_1 .

$$\text{Output signal} = r_o(t) = m_o(t) + n_o(t)$$

$$S_{mo}(f) = S_m(f)$$

and

$$S_m(f) = k \left(1 - \frac{|f|}{f_1} \right), \quad |f| \leq f_1$$

and

$$S_{no}(f) = \frac{\mathcal{N}}{2}, \quad |f| \leq f_1$$

$$\text{Therefore, output signal power} = S_o = 2k \int_0^{f_1} \left(1 - \frac{f}{f_1} \right) df = kf_1$$

The r.h.s is the area under the triangle for $S_m(f)$.

Thus, $S_o = kf_1$.

$$\text{The output noise power} = N_o = 2 \int_0^{f_1} \frac{\mathcal{N}}{2} df = \mathcal{N} f_1.$$

Hence, output SNR for case (a) is

$$(\text{SNR})_a = \frac{kf_1}{\mathcal{N}f_1} = \frac{k}{\mathcal{N}}$$

2. Solution is appended.

3. Solution

2)

$$\begin{aligned}
R_X(t + \tau, t) &= E[X(t + \tau)X(t)] = E[(X + Y)(X + Y)] \\
&= E[X^2] + E[Y^2] + E[YX] + E[XY] \\
&= E[X^2] + E[Y^2] + 2E[X]Y
\end{aligned}$$

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where the last equality follows from the independence of X and Y . But, $E[X] = 0$ since X is uniform on $[-1, 1]$ so that

$$R_X(t + \tau, t) = E[X^2] + E[Y^2] = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

The Fourier transform of $R_X(t + \tau, t)$ is the power spectral density of $X(t)$. Thus

$$S_X(f) = \mathcal{F}[R_X(t + \tau, t)] = \frac{2}{3}\delta(f)$$

$$\begin{aligned}
u(t) &= X \cos 2\pi ft - Y \sin 2\pi ft \\
E[u(t)] &= E(X) \cos 2\pi ft - E(Y) \sin 2\pi ft
\end{aligned}$$

and :

$$\begin{aligned}
R_{uu}(t, t + \tau) &= E\{[X \cos 2\pi ft - Y \sin 2\pi ft][X \cos 2\pi f(t + \tau) - Y \sin 2\pi f(t + \tau)]\} \\
&= E(X^2) [\cos 2\pi f(2t + \tau) + \cos 2\pi f\tau] + E(Y^2) [-\cos 2\pi f(2t + \tau) + \cos 2\pi f\tau] \\
&\quad - E(XY) \sin 2\pi f(2t + \tau)
\end{aligned}$$

For $u(t)$ to be wide-sense stationary, we must have : $E[u(t)] = \text{constant}$ and $R_{uu}(t, t + \tau) = R_{uu}(\tau)$. We note that if $E(X) = E(Y) = 0$, and $E(XY) = 0$ and $E(X^2) = E(Y^2)$, then the above requirements for WSS hold; hence these conditions are necessary. Conversely, if any of the above